MANIPAL
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ReceiptSense

Brief Literature Review

Topic: Document Image Analysis

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1. What is our project?

ReceiptSense is a proposed computer vision system that reads a scanned or photographed printed receipt and extracts four fields: **company name, address, date and total amount**. It will display the image regions supporting each field and let a user correct uncertain predictions before exporting a structured record.

The problem is more than reading characters. A receipt can contain subtotal, tax, cash and total amounts with similar appearances. Blur, faded printing, rotation and perspective distortion make the task harder. Our system will combine image correction, optical character recognition (OCR) and a layout-aware model to recover useful information.

The project is useful for assisted expense entry, receipt indexing and small-business record organization. Computer vision supplies the image processing, text localization, character recognition and visual-layout evidence; validation rules and the interface make those outputs usable.

Research basis: LayoutLMv3 [1] is the main model reference. SROIE [2] defines the receipt task and benchmark. Supporting papers explain text detection, recognition, spatial understanding and alternative approaches. This review describes planned implementation; project results have not yet been measured.

2. Main base paper and benchmark reference

2.1 Main base paper: LayoutLMv3 [1]

Paper: Y. Huang et al., *LayoutLMv3: Pre-training for Document AI with Unified Text and Image Masking*, ACM Multimedia, 2022.

What the paper proposes: A document transformer that combines text, its two-dimensional positions and image patches. It uses unified text/image masking and word-patch alignment during pretraining to learn relationships between written content and visual evidence.

What we use: We will adapt a pretrained LayoutLMv3 checkpoint for receipt field classification. The input will include the receipt image, OCR words and normalized bounding boxes. Predicted word labels will be grouped into company, address, date and total values.

Why it fits: Receipt meaning depends on both words and arrangement. A model can use nearby labels and positions to distinguish candidate amounts. We will fine-tune an existing model rather than recreate its large-scale pretraining. OCR errors and incorrectly aligned boxes can still affect extraction.

2.2 Benchmark reference: SROIE [2]

Paper: Z. Huang et al., *ICDAR2019 Competition on Scanned Receipt OCR and Information Extraction*, ICDAR, 2019.

What the paper proposes: A receipt benchmark with separate tasks for text localization, text recognition and key information extraction. Its field-extraction task concerns company, address, date and total.

What we use: The four-field task, available annotated receipts and the separation of OCR errors from extraction errors. We will audit the downloaded dataset release and use fixed training, validation and test partitions. Final outputs will be evaluated against held-out field labels.

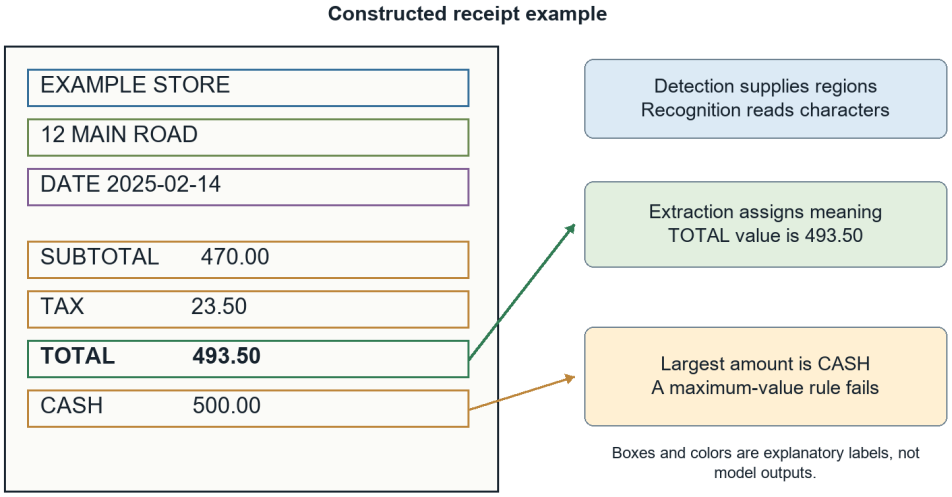


Figure 1. Reused explanatory receipt illustration: reading an amount and deciding that it is the total are different tasks. This is a constructed example, not an experimental result.

3. Supporting papers: detection and spatial understanding

3.1 DBNet: finding text regions [3]

What the paper proposes: Differentiable binarization inside a segmentation-based detector. The model learns text-region maps and adaptive thresholds so the system can locate text of different shapes.

What we use: The detection principle and the importance of preserving text-region coordinates. ReceiptSense will obtain polygons through a pretrained OCR pipeline and use them for crops and evidence overlays. We will record the actual detector checkpoint; citing DBNet does not imply that every PaddleOCR configuration implements the original DBNet architecture.

3.2 CRNN: recognizing character sequences [4]

What the paper proposes: A convolutional recurrent network combining visual feature extraction, sequence modeling and connectionist temporal classification (CTC). It recognizes variable-length text without manually segmenting each character.

What we use: The conceptual separation between detecting a text region and recognizing its characters, plus the idea of sequence-based OCR. We will initially reuse a pretrained recognizer. CRNN is a foundational reference, not a claim that we will train its architecture from scratch.

3.3 LayoutLM: combining words and positions [5]

What the paper proposes: Document pretraining that jointly models text and layout, with visual features also used to represent word appearance.

What we use: The justification for retaining bounding boxes instead of reducing a receipt to a plain text string. Its model family motivates our selected LayoutLMv3 extractor. LayoutLM itself is background literature, not an additional required model implementation.

3.4 BROS: learning spatial relationships [6]

What the paper proposes: A document model using text and relative two-dimensional positions without relying on image features. It also examines challenges from incorrect text ordering.

What we use: The importance of spatial relationships and careful reading-order reconstruction. It motivates an optional text-plus-layout comparison when resources allow. BROS will not be treated as a reproduced baseline unless we actually implement and evaluate it.

Common lesson: These papers address different stages. Detection tells us where text is, recognition tells us what it says, and spatial understanding helps decide which field it represents. Improving one stage does not automatically guarantee correct final receipt fields.

4. Alternatives and our proposed approach

4.1 TrOCR: an alternative recognizer [7]

What the paper proposes: An OCR encoder-decoder using pretrained image and text transformers to convert a text image into a character or wordpiece sequence.

What we use: A possible future comparison for recognition on suitable text crops. It helps explain transformer-based OCR, but is not required for the initial system. It recognizes text; it does not independently assign receipt fields.

4.2 Donut: an OCR-free alternative [8]

What the paper proposes: An image-to-structured-output document transformer that avoids a separate OCR engine, addressing OCR cost and error propagation.

What we use: The awareness that OCR errors can propagate into field extraction. Donut is an alternative and future extension; our initial pipeline retains explicit OCR words and regions because they are useful for inspection and correction.

4.3 Joint receipt transformer and rules [9]

What the paper proposes: Mifsud et al. study receipt extraction using a joint multimodal transformer and a rule-based model.

What we use: Related-work support for examining learned extraction alongside rules. ReceiptSense will first compare a rules baseline with LayoutLMv3 using shared OCR outputs and then apply format checks to results. We do not claim to reproduce the paper's full method or that combining rules with a transformer is a new invention.

4.4 Rough implementation idea

We will validate the receipt image, apply conservative geometry or quality correction where appropriate, and run pretrained OCR. OCR words and boxes will feed a rules baseline and the learned extractor. Both will produce the same four-field schema. A final interface will show source regions, flag ambiguous fields and support correction/export.

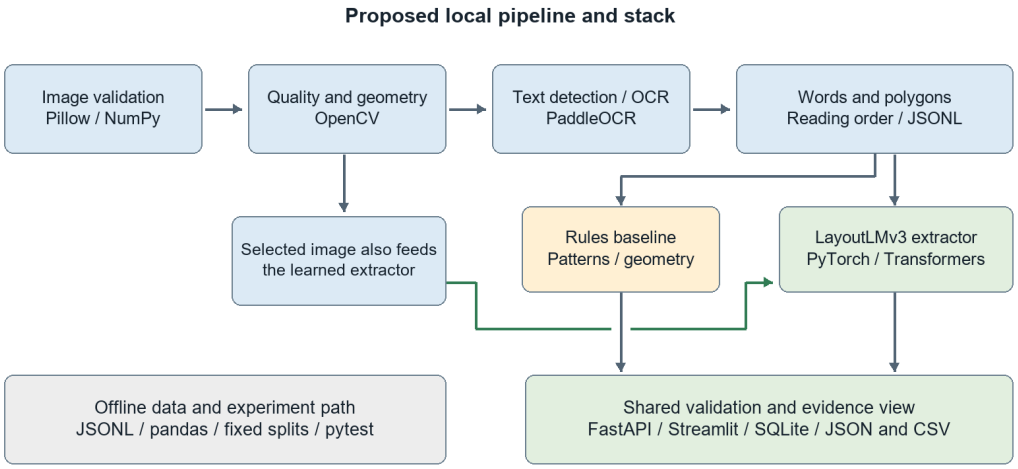


Figure 2. Reused proposed architecture. OpenCV supports image processing, PaddleOCR supplies OCR, PyTorch/Transformers support LayoutLMv3, and FastAPI/Streamlit support the local interface.

5. Literature synthesis and references

5.1 What we take into the project

Reference	Role in ReceiptSense
LayoutLMv3 [1]	Main pretrained model to adapt for four-field extraction
SROIE [2]	Receipt task, annotated benchmark and field definitions
DBNet [3] and CRNN [4]	Foundations for text detection and recognition
LayoutLM [5] and BROS [6]	Motivation for boxes, layout and spatial relationships
TrOCR [7] and Donut [8]	Optional recognition and OCR-free alternatives
Joint receipt model [9]	Related hybrid approach and a boundary on originality claims

Our proposed contribution is a practical, inspectable receipt pipeline and a controlled comparison of image correction, rules and learned extraction. We will compare exact field matches on the same held-out receipts, inspect OCR failures and test selected image degradations. Any improvement will be reported only after measurement. The literature supplies the model ideas; the team must implement their adaptation and establish whether they help our task.

References

[1] Y. Huang, T. Lv, L. Cui, Y. Lu and F. Wei. *LayoutLMv3: Pre-training for Document AI with Unified Text and Image Masking*. ACM Multimedia, 2022. https://arxiv.org/abs/2204.08387

[2] Z. Huang et al. *ICDAR2019 Competition on Scanned Receipt OCR and Information Extraction*. ICDAR, 2019; public preprint, 2021. https://arxiv.org/abs/2103.10213

[3] M. Liao, Z. Wan, C. Yao, K. Chen and X. Bai. *Real-time Scene Text Detection with Differentiable Binarization*. AAAI, 2020. https://arxiv.org/abs/1911.08947

[4] B. Shi, X. Bai and C. Yao. *An End-to-End Trainable Neural Network for Image-based Sequence Recognition and Its Application to Scene Text Recognition*. IEEE TPAMI, 2017; preprint, 2015. https://arxiv.org/abs/1507.05717

[5] Y. Xu et al. *LayoutLM: Pre-training of Text and Layout for Document Image Understanding*. KDD, 2020. https://arxiv.org/abs/1912.13318

[6] T. Hong et al. *BROS: A Pre-trained Language Model Focusing on Text and Layout for Better Key Information Extraction from Documents*. AAAI, 2022. https://arxiv.org/abs/2108.04539

[7] M. Li et al. *TrOCR: Transformer-based Optical Character Recognition with Pre-trained Models*. AAAI, 2023; preprint, 2021. https://arxiv.org/abs/2109.10282

[8] G. Kim et al. *OCR-free Document Understanding Transformer*. ECCV, 2022. https://arxiv.org/abs/2111.15664

[9] X. Mifsud, L. Grech, A. Baldacchino, L. Keller, G. Valentino and A. Muscat. *Receipt Information Extraction with Joint Multi-Modal Transformer and Rule-Based Model*. Machine Learning and Knowledge Extraction, 7(4), 167, 2025. https://doi.org/10.3390/make7040167